

Modules – Communication

Menu: Modules > Communication

Configures how the application communicates with rate control modules. Select either **Ethernet** or **CAN Bus** as the communication type — the two are mutually exclusive. The settings for each are on the **Ethernet** and **CanBus** tabs.

Option	Description
Ethernet	Modules communicate via UDP over the local network. Default mode.
CAN Bus	Modules communicate via a CAN adapter connected to this PC. Ethernet remains active for AgOpenGPS.

Ethernet

Select the PC network interface used to communicate with modules.

Control	Description
IP Address dropdown	Lists active network interfaces on this PC. Select the one on the same subnet as the modules.
Module Subnet	Shows the current subnet the application is using to reach modules.

Buttons

Button	Action
Send Subnet	Sends the selected IP subnet to the module so it knows which address to respond to
Rescan	Refreshes the IP address list and module subnet display

Module WiFi (ESP32)

ESP32 modules run their own hotspot and settings page. Connect a phone, tablet or laptop to the module's hotspot — named **RateModule_** followed by eight characters unique to that board — and browse to **192.168.200.1**, then press **Network**. Modules with an ID other than 0 use 192.168.(200 + *module ID*).1, so module 1 is at 192.168.201.1. The hotspot is open unless a hotspot password has been set.

The Network page is where the module is pointed at your own WiFi network, so it can reach the application over an existing network instead of you connecting to the module directly.

Control	Description
Network	Name of the WiFi network the module joins
Password	Password for that network
Use this Network	Ticked, the module joins the network above. Unticked, the module runs its hotspot only.
Scan for Networks	Lists the networks in range with signal strength, channel, and a padlock on encrypted ones. Tap a network to fill in its name.
Retry Connection Now	Forces an immediate connection attempt. Shown only when the module is set to join a network and is not connected.
Hotspot Password	8 to 10 characters. Leave empty for an open hotspot.
Save/Restart	Saves the settings and restarts the module

Note: Tap a network in the scan results *before* typing the password. Tapping reloads the page, which clears anything already typed.

Status

Status	Meaning
Wifi Connected to <i>name</i> (address) on channel <i>n</i>	Joined. The address shown is the module's address on your network.
Wifi Not Connected — retrying in the background	Set to join a network but not connected. The module keeps trying.
Password refused by <i>name</i>	The network rejected the password. Correct it and press Save/Restart.
Hotspot only	Use this Network is unticked.

Joining and Retrying

A module set to join a network keeps trying for as long as the setting is ticked. It does not switch itself back to hotspot-only, and it does not restart itself, so a network that goes down — the shed WiFi switched off, or the machine driving out of range — no longer interrupts work in progress. Earlier firmware gave up after several failures, turned **Use this Network** off and rebooted the module.

Attempts are deliberately spaced out. The module has a single radio shared between its hotspot and its connection to your network, so every attempt takes the hotspot briefly off the air. While a device is connected to the hotspot — a tablet running the application, for instance — attempts are about two minutes apart, or about five minutes if the module has never successfully joined that network. With nothing on the hotspot the module tries more often. Press **Retry Connection Now** rather than waiting.

Loading the module's settings pages pauses the retries for a minute after the last page, so the pages stay responsive while you are configuring. **Retry Connection Now** works regardless.

Scanning for networks takes the radio off the hotspot for a second or two, so a brief pause after pressing **Scan for Networks** is expected.

Note: The hotspot uses the same channel as the network the module joins. Devices on the hotspot drop once the first time a module joins a network, and again if that network later changes channel — they

reconnect on their own. From then on the module starts its hotspot on the remembered channel.

Some routers refuse a wrong password in a way the module cannot tell apart from a weak signal. When that happens the status stays on **Wifi Not Connected** instead of naming the password, so check the password whenever a module will not join a network you know is in range.

CAN Bus Settings

These settings apply when **CAN Bus** is selected.

CAN Adapter

Driver	Description
SLCAN	Serial port-based CAN adapter. Requires COM port selection below.
InnoMaker USB2CAN	Native USB CAN adapter (InnoMaker hardware). No COM port required.
PCAN	Native USB CAN adapter (Peak PCAN hardware). No COM port required.

COM Port (SLCAN only)

1. Use the dropdown to select the COM port the SLCAN adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

Diagnostics

Option	Description
Diagnostics	Enables CAN traffic debug logging in the activity log

Status Indicators

Indicator	Meaning
 Connected	Green — module data received within the last 2 seconds
 Driver Found	Green — CAN adapter is open and receiving frames within the last 4 seconds

Note: Changing the CAN adapter type or COM port while CAN Bus is active will restart the CAN bridge with the new settings.